Thermodynamics — 1■Hour Refresher

A compact, equation■forward review suitable for ~1 hour. Skim headings, then use the summary tables and identities at the end.

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0) Notation & Conventions
- Closed system (no mass exchange). Sign convention: heat into system = +, work by system = +.
- State functions: U, H, S, F, G, P, V, T, μ. Process quantities (not state functions): Q, W.

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1) First Law of Thermodynamics (Energy Conservation)
Statement: Energy is conserved; changes of internal energy equal heat in minus work out.

Finite form: $\Delta U = Q - W$.

Differential (reversible, simple compressible):
$dU = \delta Q - \delta W = T\,dS - P\,dV$.

Special cases:
- Isochoric ($dV=0$): $\Delta U = Q$.
- Adiabatic ($Q=0$): $\Delta U = -W$.

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2) Second Law of Thermodynamics (Direction + Entropy)
Clausius form: Heat does not flow from cold to hot without work input.
Kelvin–Planck form: No cyclic device converts all heat from a single reservoir into work.

Entropy definition (reversible): $dS = \delta Q_{rev} / T$.

Clausius inequality (any cycle): ■ $\delta Q/T \le 0$, equality only if reversible.

Isolated system: $\Delta S_{universe} = \Delta S_{sys} + \Delta S_{surr} \ge 0$.

Ideal-gas entropy changes:
$\Delta S = n C_V \ln(T2/T1) + nR \ln(V2/V1) = n C_P \ln(T2/T1) - nR \ln(P2/P1)$.

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(Additional sections from refresher included)...